

Closira AI Agent

AI customer-support workflow for **Bloom Aesthetics Clinic**, built as a focused AI Engineering project for Closira.

The agent answers SOP-grounded customer questions, qualifies high-intent leads, detects escalation triggers, and produces a structured end-of-session summary for clinic staff.

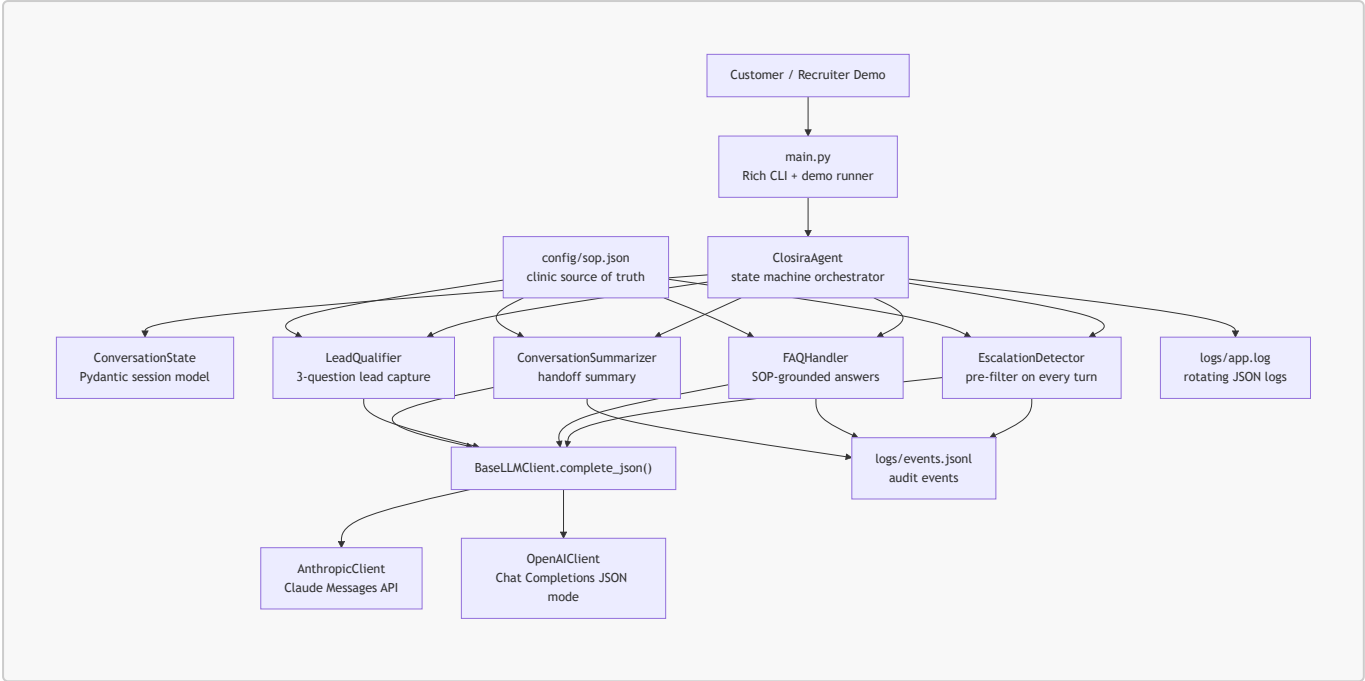
Recruiter Snapshot

Area	What this project demonstrates
AI workflow design	Multi-stage agent pipeline with explicit state transitions, deterministic routing, and stage-specific prompts
LLM integration	Provider-agnostic Anthropic/OpenAI client with structured JSON responses
Safety and reliability	Dedicated escalation pre-filter, SOP-bound answers, low-confidence handling, and audit logs
Backend engineering	Typed Pydantic state models, modular stage handlers, no global conversation state
Product thinking	Clinic-ready CLI demo, lead qualification flow, and clear handoff summaries for human coordinators

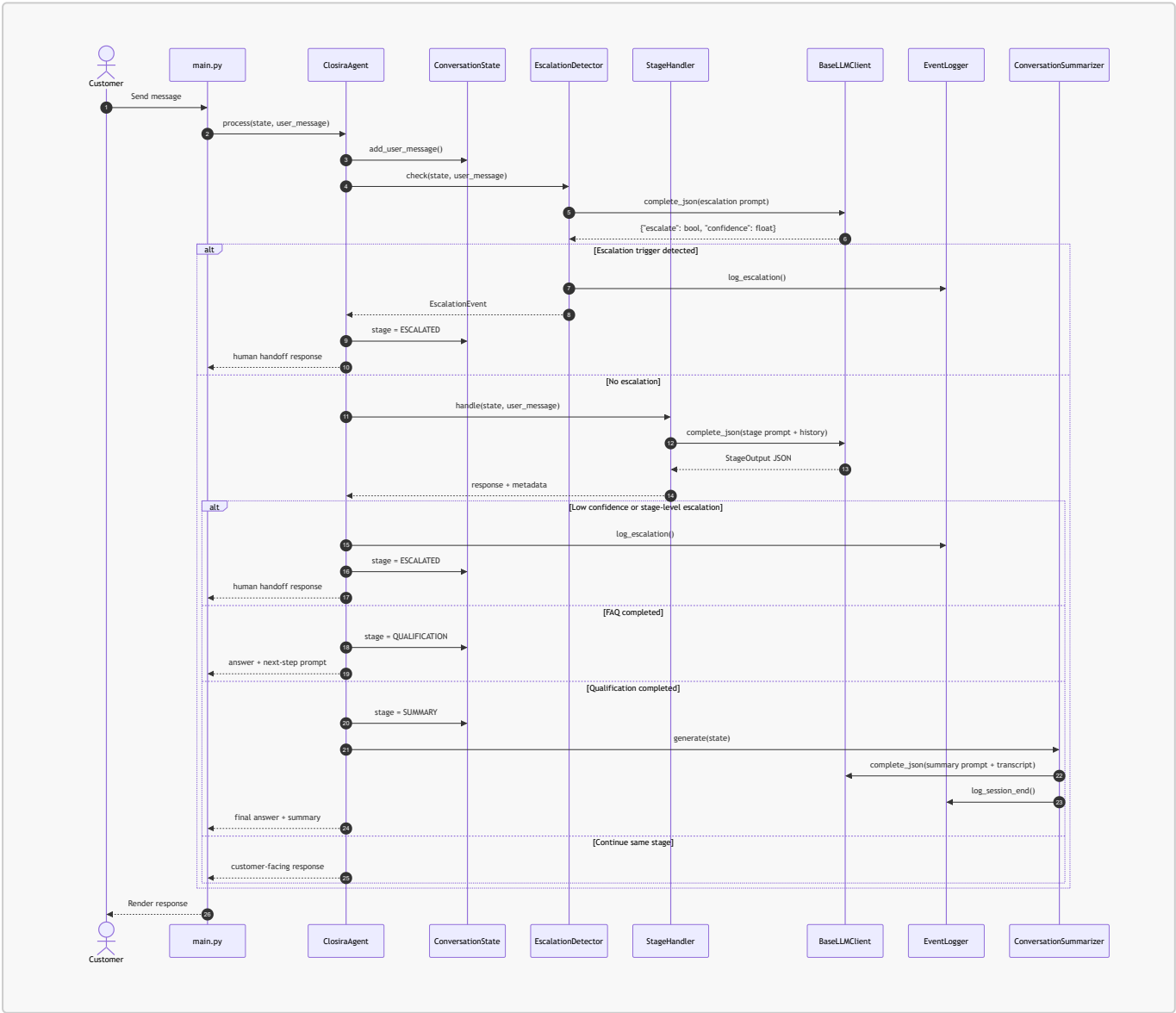
What It Does

- Answers customer FAQs using only `config/sop.json`.
- Routes every active message through an escalation classifier before normal handling.
- Qualifies interested customers with three structured lead questions.
- Escalates complaints, medical questions, pricing negotiation, safety concerns, human requests, repeated SOP gaps, and low-confidence responses.
- Generates a final `ConversationSummary` with intent, lead details, SOP gaps, escalation context, and recommended next action.
- Supports Anthropic Claude and OpenAI GPT through the same `BaseLLMClient` interface.

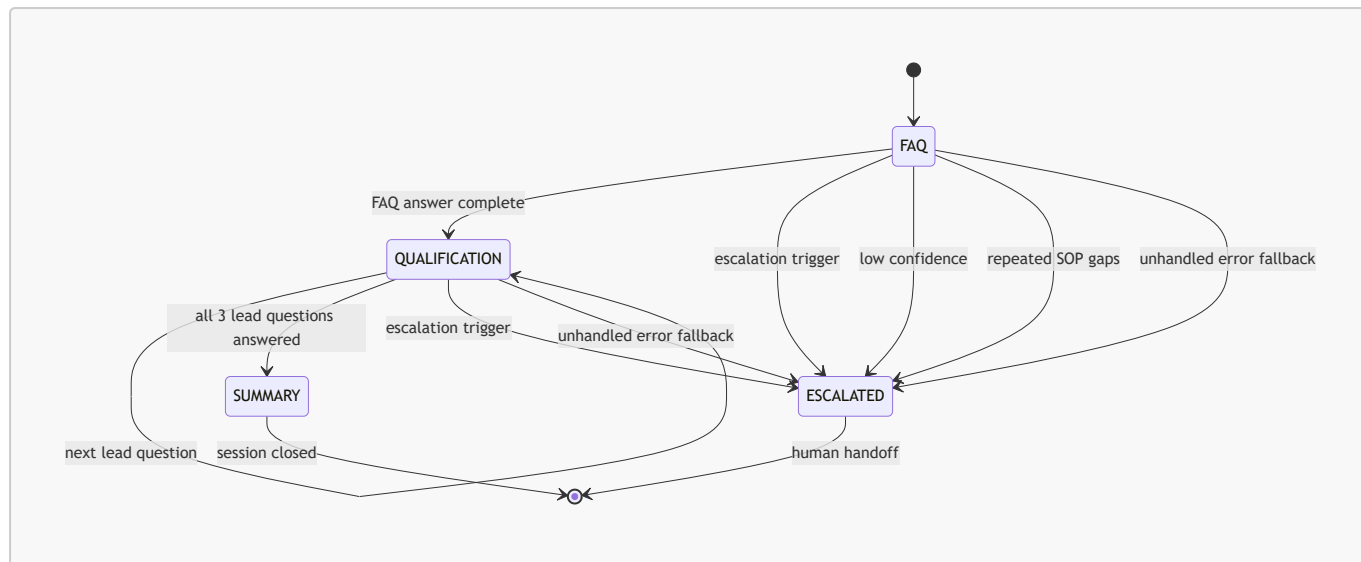
Architecture



Per-Turn Processing Pipeline

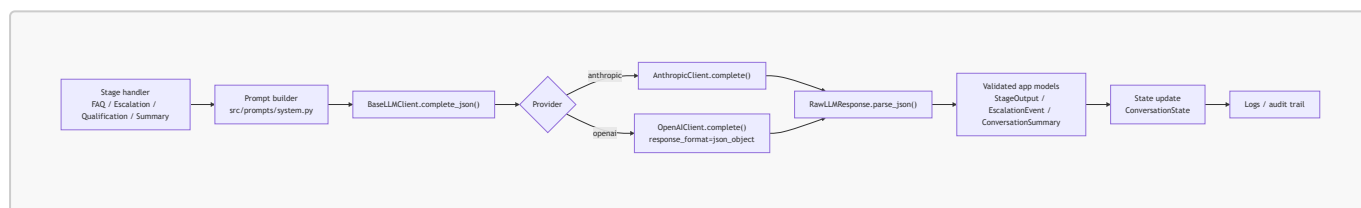


State Machine



Structured LLM Call Pipeline

The project does not depend on provider-specific native tool calling. Instead, each stage uses a **function-call-style JSON contract**: the prompt defines the exact schema, the provider client returns JSON text, `RawLLMResponse.parse_json()` extracts a valid object, and the stage converts it into typed application state.



Core JSON Contract

Every stage response follows the same predictable shape, which keeps parsing and routing simple:

```

{
  "response": "customer-facing message",
  "confidence": 0.95,
  "escalate": false,
  "escalation_reason": null,
  "escalation_type": null,
  "stage_complete": false,
  "sop_gap_detected": false,
  "sop_gap_description": null,
  "extracted_data": {}
}

```

Project Structure

```

closira-ai-agent/
|-- main.py                # CLI entry point: interactive and demo modes
|-- requirements.txt
|-- .env.example

```

```
|-- prompt_design.md          # Prompt strategy, trade-offs, and rationale
|-- config/
|   |-- sop.json              # Bloom Aesthetics Clinic SOP
|-- src/
|   |-- agent.py              # Orchestrator and state-machine routing
|   |-- models/
|       |-- schemas.py        # Pydantic models for state, events, summaries
|       |-- prompts/
|           |-- system.py      # Prompt builders and JSON output contracts
|           |-- stages/
|               |-- escalation.py  # Dedicated escalation classifier
|               |-- faq.py        # SOP-grounded FAQ answering
|               |-- qualifier.py   # Lead qualification flow
|               |-- summarizer.py  # Final conversation summary
|       |-- utils/
|           |-- llm_client.py    # Anthropic/OpenAI abstraction
|           |-- logger.py        # Structured app and audit logging
|-- test_transcripts/
|   |-- 01_in_sop_question.md
|   |-- 02_out_of_scope.md
|   |-- 03_escalation_trigger.md
|   |-- 04_lead_qualification.md
|   |-- 05_conversation_summary.md
```

Key Engineering Decisions

Decision	Why it matters
Stage-specific prompts	Each task stays narrow: answering, qualifying, escalating, or summarizing. This makes behavior easier to test and improve.
Escalation pre-filter	Safety-sensitive messages are checked before normal routing, so a stage prompt cannot accidentally continue the conversation when handoff is required.
Structured JSON outputs	LLM responses become predictable program inputs instead of brittle free text.
Provider abstraction	Anthropic and OpenAI can be swapped with <code>--provider</code> while the rest of the agent stays unchanged.
Pydantic state model	<code>ConversationState</code> is serializable, validated, and passed through every handler without hidden global state.
Audit logging	Escalations, SOP gaps, and session end events are written as JSONL for review and operations visibility.

Setup

Prerequisites

- Python 3.10+
- An Anthropic API key or an OpenAI API key

Installation

```
git clone https://github.com/Nihal108-bi/closira-ai-agent.git
cd closira-ai-agent

python -m venv .venv
source .venv/bin/activate    # Windows: .venv\Scripts\activate

pip install -r requirements.txt

cp .env.example .env
# Add ANTHROPIC_API_KEY or OPENAI_API_KEY to .env
```

Running the Agent

Interactive Mode

```
python main.py
```

Useful options:

```
python main.py --provider openai
python main.py --model claude-3-5-sonnet-20241022
python main.py --debug
```

In-session commands:

Command	Behavior
/summary	Generate a summary of the current session
/quit	End the session

Demo Mode

```
python main.py --demo
```

Demo mode runs five prebuilt scenarios:

- 1. In-SOP question
- 2. Out-of-scope question
- 3. Escalation trigger
- 4. Lead qualification
- 5. Full session with summary

SOP Grounding

The agent's source of truth is `config/sop.json`. It includes:

- Business name, location, hours, and booking channels
- Botox, dermal fillers, and consultation details
- Prices, treatment durations, expected results, and aftercare notes
- Cancellation and rescheduling policy
- Escalation trigger definitions
- Frequently asked questions

To adapt the agent for another business:

1. Replace `config/sop.json`.
2. Update the qualification questions in `src/prompts/system.py`.
3. Keep the orchestration code unchanged unless the workflow itself changes.

Escalation Rules

Trigger	Example
Negative sentiment or complaint	"I'm really unhappy with my last treatment."
Medical question	"Can I have Botox while taking medication?"
Pricing negotiation	"Can you do fillers cheaper?"
Explicit human request	"I want to speak to a real person."
Safety concern	"My face is swollen after treatment."
Repeated SOP gaps	More than two consecutive questions not covered by the SOP
Low model confidence	Confidence below 50 percent after stage processing

When a trigger fires, the agent moves to `ESCALATED`, returns the human handoff message, and records the event in `logs/events.jsonl`.

Logging and Observability

File	Purpose
<code>logs/app.log</code>	Rotating JSON application logs
<code>logs/events.jsonl</code>	Audit log for escalations, SOP gaps, and session end events

Example audit event shape:

```
{
  "ts": "2026-05-23T00:00:00Z",
  "event_type": "ESCALATION",
  "session_id": "abc12345",
  "trigger_type": "medical_question",
}
```

```
"reason": "Customer asked for medical suitability advice.",
"confidence": 0.94,
"flagged_message": "Can I have Botox while pregnant?"
}
```

Evaluation Coverage

Requirement	Implementation evidence
In-SOP answering	<code>FAQHandler</code> grounds answers in <code>config/sop.json</code> ; transcript <code>01_in_sop_question.md</code>
Out-of-scope handling	SOP gap tracking and repeated-gap escalation; transcript <code>02_out_of_scope.md</code>
Escalation trigger	Dedicated classifier in <code>src/stages/escalation.py</code> ; transcript <code>03_escalation_trigger.md</code>
Lead qualification	<code>LeadQualifier</code> stores answers in <code>ConversationState.lead_data</code> ; transcript <code>04_lead_qualification.md</code>
Conversation summary	<code>ConversationSummarizer</code> returns <code>ConversationSummary</code> ; transcript <code>05_conversation_summary.md</code>
Prompt reasoning	<code>prompt_design.md</code> documents prompt strategy and trade-offs

Trade-offs and Production Improvements

Current choice	Benefit	Production upgrade
Escalation LLM call on every turn	Clear safety boundary	Add deterministic rules for obvious triggers, then use LLM for ambiguous cases
In-memory state	Simple local demo	Persist <code>ConversationState</code> in Redis or Postgres
SOP injected into prompts	Works well for small SOPs	Use retrieval with source citations for larger knowledge bases
No streaming	Simpler CLI behavior	Stream responses for WhatsApp or web chat UX
English-only prompts and SOP	Focused assignment scope	Add language detection and localized SOP content

Quick Interview Talking Points

- I separated orchestration from stage behavior so each stage can be tested and improved independently.
- I used structured JSON contracts to make LLM output behave like typed function returns.
- I added a dedicated escalation classifier before normal response generation to reduce safety misses.
- I kept state explicit and serializable, which makes the design ready for future persistence or multi-channel deployment.
- I included audit logs so human operators can review why the AI escalated or where the SOP had gaps.

